

- It's easy to see that all 4 corners are of the same colour (white).
- On removal of 3 corners, we will be left with ~~13~~ $(13-3) = 10$ white squares and 12 black squares.
- A dominice can cover 1 white and 1 black square at a time.
- $\therefore$ 10 dominices can cover 10 white and 10 black squares.
- 2 black squares will remain to be covered. It cannot be covered by dominices.
- $\therefore$ A 5×5 checkerboard with 3 corners removed cannot be covered by dominices.

□

47. Use a proof by exhaustion to show that a tiling using dominices of a 4×4 checkerboard with opposite corners removed does not exist.

Solution:

W	B	W	B
B	W	B	W
W	B	W	B
B	W	B	W

WLOG, Let Row 1 and Column 1 be colored White. It's easy to see from the given checkerboard that Row 4, Column 4 is colored White.

- For a general 4×4 checkerboard, each row has 2 white squares and 2 black squares.
- $\therefore$ There are a total of $2 \times 4 = 8$ white and $2 \times 4 = 8$ black squares. On removal of opposite corners, we have 6 white and 8 black squares.
- A dominice can cover 1 white and 1 black square at a time.
- We shall try to exhaust all possible ways dominices can be used to cover this checkerboard.

Method 1

- Dominice 1: Row 2 Column 1, Row 3 Column 1. (Suppose)
- There are no more rows after Row 4. $\therefore$ Dominice 2 must be applied on Row 4 Column 1 and Row 4 Column 2. Row 4 Column 4 is removed.
- Dominice 3: must be applied on Row 4 Column 3 and Row 3 Column 3.
- Again, Dominice 4: must be applied on Row 3 Column 4 and Row 2 Column 4.